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Submitter Information

Email: [REDACTED]
Organization: SAFE Center for Grid Security

General Comment

See attached file(s)

Attachments

SAFE CGS National AI Action Plan Comments Final

To: National Science Foundation (NSF)

2415 Eisenhower Avenue
Alexandria, VA 22314

Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO)

2415 Eisenhower Avenue
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From: Danielle Russo, Executive Director, Center for Grid Security, SAFE

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Date: 03.10.2025

Pursuant to the National Science Foundation's (NSF) *"Request for Information on the Development of an Artificial Intelligence (AI) Action Plan,"*¹ SAFE provides the following comments.

I. COMMUNICATIONS

All correspondence and communication concerning this filing should be directed to:

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II. INTRODUCTION TO SAFE

SAFE is an action-oriented, nonpartisan organization committed to transportation, energy, and supply chain policies that advance the economic and national security of the United States, its partners, and allies. SAFE's Center for Grid Security (CGS) provides policy analysis and recommendations to build out the national power grid—ensuring America's energy and national security. The Center actively engages federal- and state-level policymakers, the Department of Defense (DoD), and other key stakeholders to ensure grid modernization and expansion prioritizes American national interests.

III. COMMENTS

¹ National Science Foundation, *"Request for Information on the Development of an Artificial Intelligence (AI) Action Plan,"* Document No. 2025-02305, February 6, 2025.

SAFE's CGS applauds the Trump Administration's commitment to securing U.S. leadership in artificial intelligence (AI) through a National AI Action Plan. As AI continues to shape America's technological and economic future, it is crucial to ensure that the nation's electric grid has the resilience and capacity to sustain this innovation to protect our economic and national security.

Several recent Executive Orders from the Administration underscore the strategic imperative of meeting the rapidly rising demand for electricity in support of AI expansion. However, there is a critical mismatch between the speed of the AI industry's expansion and the ability of the grid to supply resilient, reliable, and affordable power. While new AI facilities can be constructed in one to two years, bringing new energy generation online to support such facilities can take more than four years, and building necessary transmission infrastructure takes even longer. Without urgent action, America's energy infrastructure will struggle to keep pace with this and other industries that define our future competitiveness.

In support of a National AI Action Plan, SAFE's CGS offers the attached report, titled "Wired for Defense: The National Security Imperative of Transmission Expansion," which studies the throughline connecting domestic generation and transmission expansion needs, AI growth for national security, and U.S. military capability.² SAFE also strongly encourages the Administration to consider the following key priorities in a National AI Action Plan:

- **Align transmission planning with AI development:** Ensure that transmission expansion plans are synchronized with the projected growth of AI facilities to prevent bottlenecks and support seamless integration. Accelerate permitting and investment in transmission projects to ensure the grid can accommodate the rapid growth in electricity demand driven by AI and other strategic industries.
- **Prioritize AI for national defense energy systems:** Prioritize energy projects based on their support of AI applications that enhance the resilience and operational efficiency of military energy infrastructure, reducing vulnerabilities and ensuring mission readiness.
- **Implement grid-enhancing technologies (GETs):** Adopt technologies that improve the efficiency and capacity of existing transmission lines, offering faster solutions to increasing grid capacity without the need for new construction.
- **Scale grid-level capacity for strategic industries:** AI innovation and manufacturing growth require significant new investments in electricity generation and transmission. A more cost-effective approach to ensuring reliability is rapidly expanding interregional transmission through permitting reform, which can deliver power at a fraction of the cost of new local generation and reduce the risk of grid bottlenecks.³

² Danielle Russo, "Wired for Defense: The National Security Imperative of Transmission Expansion," SAFE Center for Grid Security, November 22, 2024.

³ John D. Wilson, Zach Zimmerman, and Rob Gramlich, "Strategic Industries Surging: Driving US Power Demand," Grid Strategies, December 2024, at 6.

- **Mitigate supply chain risks by boosting domestic manufacturing for grid power and AI:** AI-driven infrastructure must be protected from adversarial threats, including cyber intrusions and supply chain vulnerabilities, particularly those linked to foreign-controlled technologies. Identifying and securing the energy needs to boost domestic manufacturing of such infrastructure will be needed to fully protect America's AI posture.

SAFE looks forward to the publishing of a National AI Action Plan that strengthens both America's economic and national security while safeguarding the resilience of the U.S. energy grid.

Respectfully Submitted,

/s/ Danielle Russo

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